

Revenue Maximization in Auction Problems

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ABSTRACT

We study the problem of revenue maximization in single-item auctions where auction information is disseminated through a directed social network. In this setting, a seller initiates the auction and relies on strategic agents embedded in the network to propagate auction awareness to their neighbors. Each agent privately decides not only how much to bid but also whether to forward the auction to others, making the information diffusion process endogenous and strategic. We formulate the problem of designing a diffusion auction mechanism that is both Bayesian incentive compatible and individually rational, and maximizes the seller's expected revenue. To tackle this, we propose a linear programming framework that computes optimal allocation and payment rules over a known network structure and discrete valuation distributions. We validate our framework by recovering the Myerson optimal auction in a baseline setting, and extend it to more complex networked scenarios. Simulations for small networks (with two and three agents) provide insights into how optimal mechanisms adapt to network structure and diffusion constraints, highlighting novel interactions between information spread and revenue optimization.

KEYWORDS

Single Indivisible Good Auctions, Revenue Maximization

1 INTRODUCTION

We examine a setting where a single indivisible good is sold through an auction mechanism operating over a network with directed information flow. A seller initiates the process, and multiple agents embedded in the network act as potential buyers. The dissemination of auction information depends on the willingness of intermediate agents to forward the message, thereby making information propagation a strategic decision. This leads to a game-theoretic environment where agents must decide not only how much to report as their value for the item but also whether to share the auction's existence with their neighbors. The interaction blends elements of mechanism design with the structure and incentives found in networked communication. Our aim as the mechanism designer is to incentivize all agents to truthfully reveal their valuation, and to propagate auction information to all neighbors.

This problem arises in several real-world scenarios, for instance, social media auctions for limited products like concert tickets, where a seller promotes the auction to selected influencers, who then decide whether to share the opportunity with their followers. The underlying information flow can be modeled as a directed

graph, where nodes represent individuals and directed edges indicate who can inform whom, capturing the asymmetric nature of influence and communication in such networks.

Our objective is to formulate a computational method of finding the revenue maximizing mechanism for any given network, where we have information about the network structure and the agent valuations. We frame it as an optimization problem and propose a framework to find the optimal solution, and give insights on how the optimal mechanism is structured.

1.1 Related Work

Li et al. [2] study mechanism design in the context of social networks, proposing incentive-compatible mechanisms that account for network structure. However the mechanisms proposed have been later proven to be not truthful or poor revenue raisers.

2 MODEL

2.1 Notation and Assumptions

The system is modeled as a directed graph $G = (N \cup \{s\}, E)$, where:

- $N = \{1, 2, \dots, n\}$ denotes the set of potential buyers.
- s is a special node representing the seller, who initiates the auction.
- $E \subseteq (N \cup \{s\}) \times N$ is the set of directed edges indicating potential information flow. A directed edge $(i, j) \in E$ means that node i can share information with node j .

Information about the auction propagates through the graph along directed paths, but only if each intermediate node chooses to transmit the information. Thus, a node $i \in N$ learns about the auction only if there exists a directed path from s to i such that all intermediate agents decide to share the message.

Each agent $i \in N$ is characterized by a private type $\theta_i = (v_i, r_i)$, where:

- $v_i \in V_i \subseteq [0, b_i]$ is the agent's valuation for the good.
- $r_i \subseteq N$ is the set of directed neighbors in the graph to whom agent i can forward the auction information.

The set of possible types for agent i is $\Theta_i = V_i \times R_i$, and the global type space is $\Theta = \prod_{i \in N} \Theta_i$.

Agents report their types to the mechanism as $\hat{\theta}_i = (\hat{v}_i, \hat{r}_i)$, where:

- $\hat{v}_i$ is the reported valuation.
- $\hat{r}_i \subseteq r_i$ is the subset of neighbors the agent actually forwards auction information to.

The reported type profile of all agents except i is denoted $\hat{\theta}_{-i} = (\hat{\theta}_1, \dots, \hat{\theta}_{i-1}, \hat{\theta}_{i+1}, \dots, \hat{\theta}_n)$.

We define a filtering function $\text{fil}(\hat{\theta}_i, \hat{\theta}_{-i})$, which returns the type profile restricted to the set of agents reachable from the seller s under the reported profile. This function captures the effective

subnetwork over which the auction information has actually been transmitted.

The seller s is assumed to be non-strategic and always shares the auction information with its outgoing neighbors.

We assume that the prior distribution over (θ_i, θ_{-i}) is given by P and is common knowledge. We also assume that the distributions of v_i and p_i are independent of each other, and independent across agents. Formally, we define f_i and h_i to be the distributions of valuations v_i and neighbor set r_i for each $i \in N$. Then we define $P(\theta_i, \theta_{-i}) = \prod_{i \in N} f_i(v_i) h_i(r_i)$.

2.2 Definitions

Definition 2.1. Allocation function $g : \Theta \rightarrow \Delta_n$ is defined such that its i^{th} component $g_i(\text{fil}(\theta))$ denotes the probability of agent i being allocated the object. Here we define $\Delta_n := \{x \in \mathbb{R}_{\geq 0}^n : \sum_{i=1}^n x_i = 1\}$ i.e., the space of all possible allocation vectors.

Definition 2.2. Payment function $p : \Theta \rightarrow \mathbb{R}$ is defined such that its i^{th} component p_i denotes the payment that agent i must make.

The defined allocation and payment functions allow us to formally define our diffusion auction (DA).

Definition 2.3. A diffusion auction (DA) is defined by the tuple (g, p) where g and p are the allocation and payment functions respectively.

Since the agents' types now contain both valuation v_i and neighbor set r_i , we need to redefine the utility function according to the subnetwork $\text{fil}(\hat{\theta}_i, \hat{\theta}_{-i})$ that remains connected after the agents choose action $\hat{r}$.

Definition 2.4. Utility function for agent i under allocation function g and payment function p is defined by the standard quasi-linear model, as

$$u_i^{(g,p)}((\hat{\theta}_i, \hat{\theta}_{-i}); \theta_i) = v_i g_i(\text{fil}(\hat{\theta}_i, \hat{\theta}_{-i})) - p_i(\text{fil}(\hat{\theta}_i, \hat{\theta}_{-i})).$$

For ease of representation, we define the expected allocation and payments as follows.

Definition 2.5. Expected allocation function $\alpha_i(v_i, r_i) = \mathbb{E}[g_i(\text{fil}(\theta_i, \theta_{-i})) | \theta_i = (v_i, r_i)]$ is obtained by taking the expectation of a particular agent's allocation over the types of all other agents (θ_{-i}) .

Definition 2.6. Expected payment function $\text{pay}_i(v_i, r_i) = \mathbb{E}[p_i(\text{fil}(\theta_i, \theta_{-i})) | \theta_i = (v_i, r_i)]$ is obtained by taking the expectation of a particular agent's payment over the types of all other agents (θ_{-i}) .

Definition 2.7. (Diffusion Bayesian Incentive Compatibility): A DA (g, p) on directed graph G is diffusion Bayesian incentive compatible (DBIC) if every agent can maximize their utility by reporting their true type, irrespective of what other agents report, i.e.,

$$v_i \cdot \alpha_i(v_i, r_i) - \text{pay}_i(v_i, r_i) \geq v_i \cdot \alpha_i(v'_i, r'_i) - \text{pay}_i(v'_i, r'_i) \\ \forall i \in N, \forall \theta_i = (v_i, r_i), \forall v'_i, r'_i \subseteq r_i$$

Definition 2.8. (Interim Individual Rationality): A DA (g, p) on directed graph G is IIR if $v_i \cdot \alpha_i(v_i, r_i) - \text{pay}_i(v_i, r_i) \geq 0$, $\forall v_i, r_i, \forall i \in N$.

The above captures the principle that each agent should expect a non-negative utility from participating in the auction, given their own type and before knowing the types of others. Note that $\text{pay}_i(0, r_i) \leq 0, \forall i \in N$ is a necessary condition for IIR.

2.3 Objective Function

Our goal is to maximize the expected revenue of the seller over the payment and allocation function. The expected revenue is given by

$$\sum_i \sum_{r_i} \int_0^{b_i} f_i(v_i) h_i(r_i) \text{pay}_i(v_i, r_i) dv_i$$

However, our linear program solver works only for discrete optimization problems, so we discretize the objective function as

$$\sum_i \sum_{r_i} \sum_{v_i=0}^{b_i} f_i(v_i) h_i(r_i) \text{pay}_i(v_i, r_i)$$

Here since we are maximizing the objective over a fixed graph structure, we neglect the $\sum_{r_i}$ term and treat $h_i(r_i)$ as a Dirac delta function that is 1 at the r_i of our graph and 0 everywhere else. Our final objective function is thus given by

$$\sum_i \sum_{v_i=0}^{b_i} f_i(v_i) \text{pay}_i(v_i, r_i)$$

where r_i denotes the set of neighbors of agent i in the graph.

2.4 Constraints

First we have the DBIC constraint

$$v_i \cdot \alpha_i(v_i, r_i) - \text{pay}_i(v_i, r_i) \geq v_i \cdot \alpha_i(v'_i, r'_i) - \text{pay}_i(v'_i, r'_i) \\ \forall i \in N, \forall (v_i, r_i), \forall v'_i, r'_i \subseteq r_i$$

This is split into the following two conditions:

$$v_i \cdot \alpha_i(v_i, r_i) - \text{pay}_i(v_i, r_i) \geq v_i \cdot \alpha_i(v'_i, r_i) - \text{pay}_i(v'_i, r_i) \\ \forall i \in N, \forall (v_i, r_i), \forall v'_i$$

$$v_i \cdot \alpha_i(v_i, r_i) - \text{pay}_i(v_i, r_i) \geq v_i \cdot \alpha_i(v_i, r'_i) - \text{pay}_i(v_i, r'_i) \\ \forall i \in N, \forall (v_i, r_i), r'_i \subseteq r_i$$

Then from the IIR condition we get

$$\text{pay}_i(0, r_i) \leq 0, \forall i, r_i$$

Since we want our expected allocation to be monotonic, we have

$$\alpha_i(v_i + 1, r_i) \geq \alpha_i(v_i, r_i), \forall i \in N, \forall v_i$$

To ensure allocation function g is a valid probability distribution, we have the constraint

$$\sum_{i=1}^n g_i(\theta) \leq 1$$

We also introduce an "Informed agents constraint", that ensures that uninformed agents (agents who never receive information about the auction) cannot receive the object, and do not make any payment. For every uninformed agent i we have

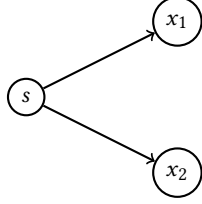
$$g_i(\theta) = 0, p_i(\theta) = 0$$

3 EXPERIMENTS AND RESULTS

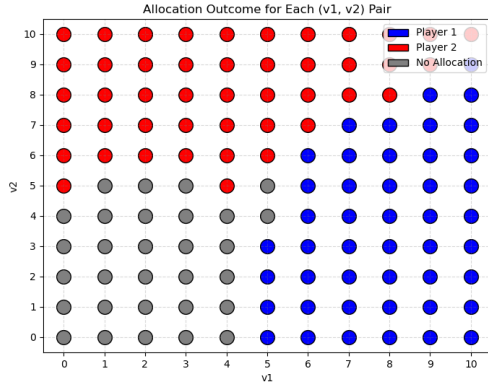
All the following simulations were run using the Gurobi linear program solver. The code and results of our experiments can be found at [1].

Since the number of constraints rises exponentially with the number of agents, we run simulations only for $n = 2$ and $n = 3$, and derive insights into the mechanism design from the results.

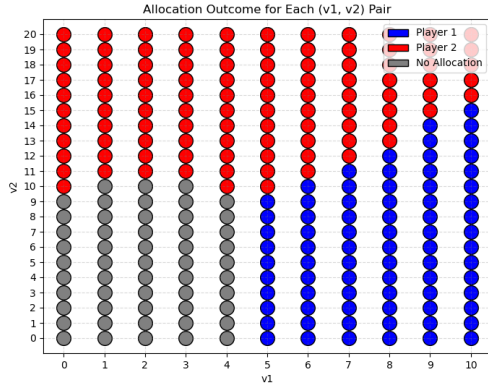
3.1 $n=2$



We implement the Myerson auction mechanism on the graph shown above, where the seller is directly connected to two agents.



(a) $b_1 = b_2 = 10$



(b) $b_1 = 10, b_2 = 20$

Figure 1: Allocations for Myerson Auction under different bids

The allocation for the Myerson Auction obtained through our implementation closely matches the theoretical predictions. Myerson's optimal auction theory provides a revenue-maximizing mechanism for single-item auctions by awarding the item to the bidder with the highest non-negative *virtual valuation*. When bidder valuations are drawn independently from uniform distributions over $[0, b_i]$, the virtual valuation for bidder i becomes $\phi_i(v_i) = 2v_i - b_i$. The Myerson mechanism allocates the item to the bidder with the highest such virtual valuation, provided it is non-negative, and charges payments that ensure incentive compatibility. In our simulation, we discretize the valuation space and allocate based on this rule. The resulting allocation plots match the theoretical boundaries implied by the virtual valuation comparison, thereby validating our implementation of the Myerson auction.

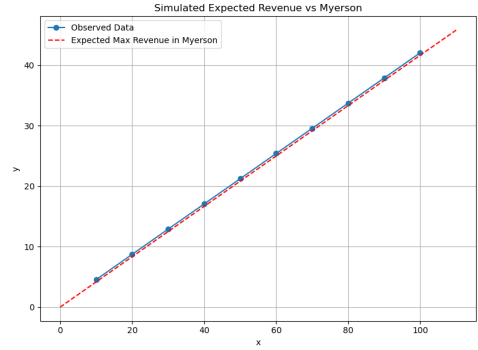
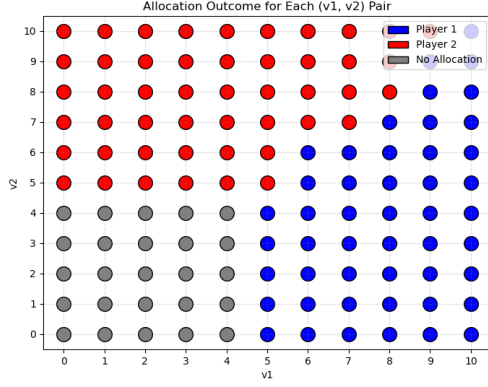


Figure 2: Expected Revenue in Linear Solver vs Myerson

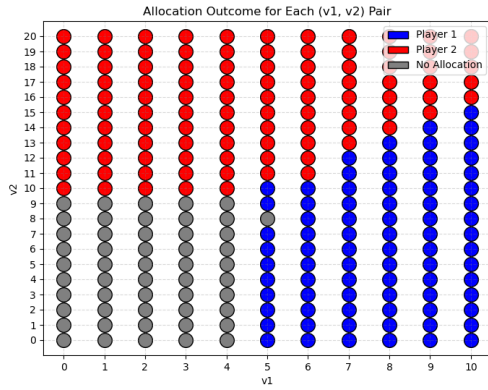
We observe that the maximum expected revenue when $b_1 = b_2 = b$ closely aligns with the theoretical prediction from Myerson's auction framework, yielding an expected revenue of $\frac{5}{12}b$ under the uniform distribution assumption, further validating our implementation of the Myerson auction.

The outcomes are then used to validate the results produced by our linear program solver, which operates on a linear chain network consisting of a seller and two agents, illustrated in the graph below.





(a) $b_1 = b_2 = 10$



(b) $b_1 = 10, b_2 = 20$

Figure 3: Allocations for Linear Network Auction under different bids

In addition to the direct auction setup, we also implemented a linear network auction model where the seller is initially connected only to player 1, who can then choose to relay the auction information to player 2. Despite this informational asymmetry, we observed that the optimal allocation rule that maximizes the seller's expected revenue remains identical to that of the classical Myerson auction. Specifically, even with the network constraint, the allocation is determined by comparing the virtual valuations of the bidders, and the item is awarded to the bidder with the highest non-negative virtual valuation. Our simulation results further show that the expected revenue obtained in this networked setting matches that of the standard Myerson auction, suggesting that the revenue-maximizing allocation structure is robust to this simple form of limited information diffusion.

3.2 $n=3$

We consider all possible graphs with a seller and 3 agents. To analyse results here, we plot the 3D allocation graphs similar to the ones plotted in the $n = 2$ case, and observe how the allocation varies over

all possible values of v_1 and v_2 while keeping v_3 fixed at certain values (by taking slices of the 3D graph).

3.2.1 Myerson Setup.

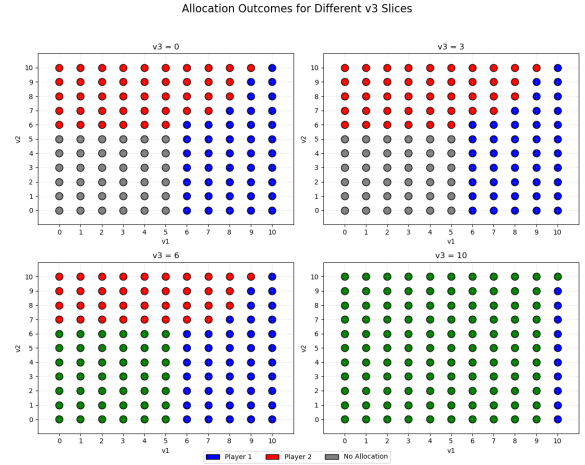
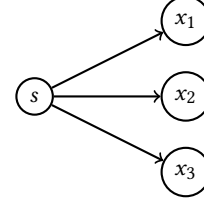


Figure 4: Allocations for Myerson setup under different bids

3.2.2 Line Graph.

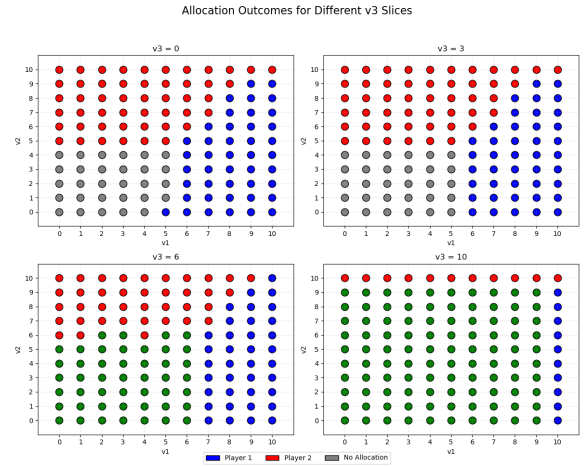
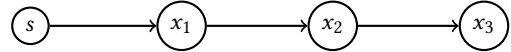


Figure 5: Allocations for Line Graph under different bids

3.2.3 Y-Graph.

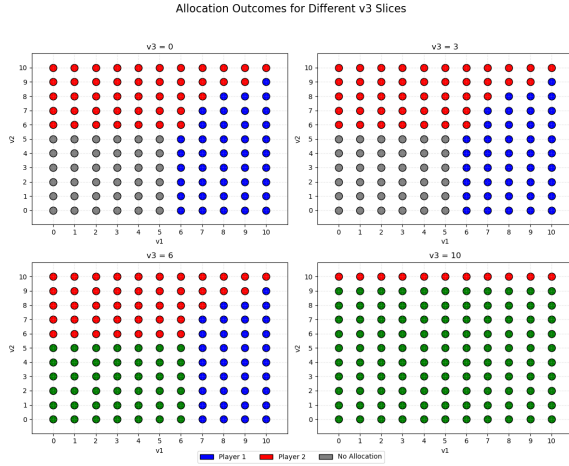
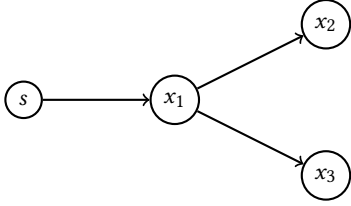


Figure 6: Allocations for Y-Graph under different bids

3.2.4 Forked Line Graph.

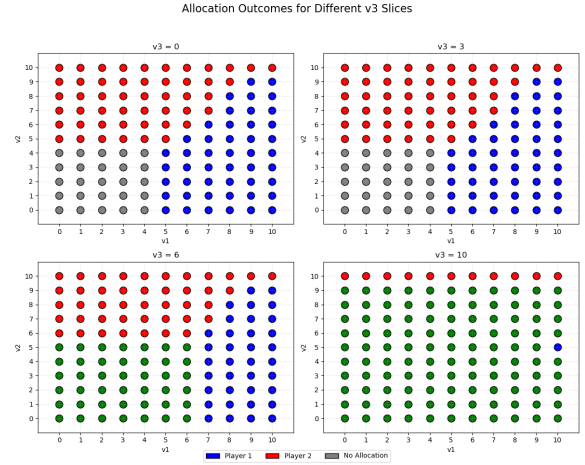
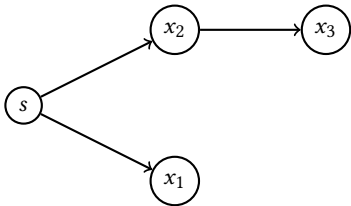


Figure 7: Allocations for Forked Line Graph setup under different bids

We observe that in each of the graphs, the allocation is exactly the same as that given by the Myerson auction (barring the corner cases). This implies that even when nodes are not connected directly to the seller we get the same allocation, meaning all agents propagate information to all their neighbors as desired. We also observe the values of the objective function in both our linear program setup and Myerson's, and find that they are nearly the same, meaning we also achieve maximum possible revenue.

We also observe that in the cases of $v_3 = 0$ and $v_3 = 3$, agent 3 is never allocated the object. This matches with our theoretical expectation, since $v_3 < b_3/2$ here. As v_3 increases (as seen in the $v_3 = 10$ case), the fraction of points where v_3 is allocated the object increases as expected.

4 CONCLUSIONS

The above results of comparing our linear program allocation with the Myerson allocation confirm that our framework achieves the maximum possible revenue while making sure that all agents propagate information to all their neighbors in the graph. The method can be further extended to setups with more agents, with optimization techniques to overcome the computational barriers.

REFERENCES

- [1] [n.d.]. Link to code and results. https://github.com/deeksha-4/cs6002_project.git.
- [2] Bin Li, Dong Hao, Dengji Zhao, and Tao Zhou. 2017. Mechanism Design in Social Networks. In *Proceedings of the 26th International Joint Conference on Artificial Intelligence (IJCAI)*.